

## GMP252

### A.2.1 Measurement Data

Table 52 Modbus Measurement Data Registers (Read-Only)

| Register Number (Decimal) | Address (Hexadecimal) | Register Description           | Data Format           | Unit                   |
|---------------------------|-----------------------|--------------------------------|-----------------------|------------------------|
| 1                         | 0000 <sub>hex</sub>   | Measured CO <sub>2</sub> value | 32-bit float          | ppm                    |
| 3                         | 0002 <sub>hex</sub>   | Compensation T                 | 32-bit float          | °C                     |
| 5                         | 0004 <sub>hex</sub>   | Measured T                     | 32-bit float          | °C                     |
| 257                       | 0100 <sub>hex</sub>   | Measured CO <sub>2</sub> value | 16-bit signed integer | ppm (up to 32 000 ppm) |

| Register Number (Decimal) | Address (Hexadecimal) | Register Description           | Data Format           | Unit                                                  |
|---------------------------|-----------------------|--------------------------------|-----------------------|-------------------------------------------------------|
| 258                       | 0101 <sub>hex</sub>   | Measured CO <sub>2</sub> value | 16-bit signed integer | ppm <sup>1)</sup> (scaled, up to approx. 320 000 ppm) |

1) The ppm output of the second Measured CO<sub>2</sub> value register (number 258) is scaled and must be multiplied by 10.

### A.2.3 Status Registers

Table 54 Modbus Status Registers (Read-Only)

| Register Number (Decimal) | Address (Hexadecimal) | Register Description   | Data Format | Notes                                                                                                                                                                 |
|---------------------------|-----------------------|------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2049                      | 0800 <sub>hex</sub>   | Device status          | 16-bit      | 0 = Status OK.<br>1 = Critical error.<br>2 = Error.<br>4 = Warning.                                                                                                   |
| 2050                      | 0801 <sub>hex</sub>   | CO <sub>2</sub> status | 16-bit      | 0 = Status OK.<br>2 = CO <sub>2</sub> reading not reliable. Appears during transmitter start-up.<br>256 = Measurement not ready. Appears during transmitter start-up. |



Multiple statuses can be present simultaneously. In those cases, the value of the status register is the sum of the status values. For example, the value of the device status register is **6** if a warning (**4**) and an error (**2**) are present simultaneously.

# HMP60

## Modbus registers ⓘ

Registers are numbered in decimal, starting from 1. Register addresses in actual Modbus messages (Modbus Protocol Data Unit (PDU)) are in hexadecimal and start from zero. Register number 1 corresponds to address 0<sub>hex</sub> in the actual Modbus message.

**CAUTION!** Reading the wrong register(s) may result in correct-looking values. Check the reference documentation of your Modbus host (PLC) to verify which notation it uses for Modbus register addresses.

## Measurement data registers ⓘ

Table 1. Modbus measurement data registers (read-only)

| Register number       | Address             | Register description        | Data format    | Unit                  |
|-----------------------|---------------------|-----------------------------|----------------|-----------------------|
| Floating point values |                     |                             |                |                       |
| 1                     | 0000 <sub>hex</sub> | Relative humidity           | 32-bit float   | %RH                   |
|                       | 0001 <sub>hex</sub> |                             |                |                       |
| 3                     | 0002 <sub>hex</sub> | Temperature <sup>1</sup>    | 32-bit float   | °C                    |
|                       | 0003 <sub>hex</sub> |                             |                |                       |
| 9                     | 0008 <sub>hex</sub> | Dew/frost point temperature | 32-bit float   | °C                    |
|                       | 0009 <sub>hex</sub> |                             |                |                       |
| 15                    | 000E <sub>hex</sub> | Absolute humidity           | 32-bit float   | g/m <sup>3</sup>      |
|                       | 000F <sub>hex</sub> |                             |                |                       |
| 17                    | 0010 <sub>hex</sub> | Mixing ratio                | 32-bit float   | g/kg                  |
|                       | 0011 <sub>hex</sub> |                             |                |                       |
| 19                    | 0012 <sub>hex</sub> | Wet-bulb temperature        | 32-bit float   | °C                    |
|                       | 0013 <sub>hex</sub> |                             |                |                       |
| 27                    | 001A <sub>hex</sub> | Enthalpy                    | 32-bit float   | kJ/kg                 |
|                       | 001B <sub>hex</sub> |                             |                |                       |
| Integer values        |                     |                             |                |                       |
| 257                   | 0100 <sub>hex</sub> | Relative humidity           | 16-bit integer | %RH * 10              |
| 258                   | 0101 <sub>hex</sub> | Temperature <sup>1</sup>    | 16-bit integer | °C * 10               |
| 261                   | 0104 <sub>hex</sub> | Dew/frost point temperature | 16-bit integer | °C * 10               |
| 264                   | 0107 <sub>hex</sub> | Absolute humidity           | 16-bit integer | g/m <sup>3</sup> * 10 |
| 265                   | 0108 <sub>hex</sub> | Mixing ratio                | 16-bit integer | g/kg * 10             |
| 266                   | 0109 <sub>hex</sub> | Wet-bulb temperature        | 16-bit integer | °C * 10               |
| 270                   | 010D <sub>hex</sub> | Enthalpy                    | 16-bit integer | kJ/kg * 10            |

<sup>1</sup> Only temperature output is available in probe model HMP110T.

Status registers 

Table 1. Modbus status data registers (read-only)

| Register number | Address                                    | Register description | Data format    | Note                                                                                                                                                                  |
|-----------------|--------------------------------------------|----------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 513             | 0200 <sub>hex</sub>                        | Error status         | 16-bit integer | 0000 <sub>hex</sub> : One or more errors active<br>0001 <sub>hex</sub> : No errors                                                                                    |
| 516             | 0203 <sub>hex</sub><br>0204 <sub>hex</sub> | Error code           | 32-bit integer |                                                                                                                                                                       |
| 518             | 0205 <sub>hex</sub><br>0206 <sub>hex</sub> | Security hash        | 32-bit integer | Security hash changes when any change is made to device settings or adjustments, but also returns back to the previous value if such changes are reverted completely. |

Table 2. Error codes on Modbus interface

| Error code register value | Corresponding error                               |
|---------------------------|---------------------------------------------------|
| 1                         | Temperature measurement error. [44]               |
| 2                         | Humidity measurement error. [45]                  |
| 4                         | Humidity sensor failure. [46]                     |
| 8                         | Capacitance reference error. [47]                 |
| 16                        | Ambient temperature out of range. [48]            |
| 32                        | Firmware checksum mismatch. [49]                  |
| 64                        | Device settings corrupted. [50]                   |
| 128                       | Additional configuration settings corrupted. [51] |
| 256                       | Sensor coefficients corrupted. [52]               |
| 512                       | Main configuration settings corrupted. [53]       |
| 2048                      | Supply voltage out of range. [55]                 |
| 8192                      | Non-volatile memory read/write failure. [57]      |
| 16384                     | Calibration certificate checksum mismatch. [58]   |

If several errors are active at the same time, the Modbus error code register contains the sum of currently active error codes.

Refer to [Error messages in Insight software](#) for more information about the error states.